Fig. 3.3 explains how to obtain ***matrix representation*** of a relation, say,

R= {(1, a), (1, b), (1, c), (2, b), (2, c), (2, d)} from the set X= {1, 2, 3, 4} to the set Y = {a, b, c, d}

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \left\{ \begin{array}{cccc} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right\} \end{array}$$

Fig. 3.3: Matrix Representation of a Relation

For representing a relation R **on** a set X**,** i. e. when domain of R = codomain of R, a directed graph may also

be used. For example, the relation R = {(1, 1), (1, 2), (2, 2), (3, 1), (3, 3)} on the set X= {1, 2, 3}, may be

represented by the following directed graph (Fig. 3.4).

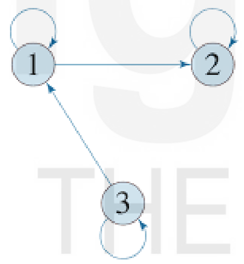

Fig 3.4: Directed Graph Representing a Relation R on a Set X

Let us recall the difference between a relation and an operation (covered in Unit 1). A ***relation*** returns a value TRUE or FALSE. For example: 3< 5 returns TRUE; 7< 5 returns FALSE. On the other hand, *an **operation** gives something new;* in the following cases new numbers: 8+15 gives new number 23; and (+√**16)** gives a new number 4. In the next subsection, we discuss relations on (set of) relations (rather, we will discuss mainly properties (i. e. unary relations) of relations. In the subsequent subsection, we discuss operations on (set of) relations.

3.2.2 Properties of Relations

We will discuss properties of only for those relations R for which the domain and range is the same set, say X, i.e., the ***relations* on** a set, say X. Unless mentioned otherwise, all relations are on the set X (fixed for all the relations under consideration). The discussion revolves around three important properties of relations on a set, viz. ***(i) reflexive, (ii) symmetric,*** and ***(iii) transitive***. Other relations to be discussed are directly or indirectly modifications of these properties.

1) **Reflexive, Not-Reflexive, Anti-Reflexive/Irreflexive**

a) *A relation R on set X is said to be **reflexive**, if **for every** a ∈ X, then (a, a) ∈ R. For example, each of the following is a reflexive relation*

i) the relation of equality (=) on N = {1, 2, 3, …}

ii) the relation of *is-sibling-of* on the set of all human beings

iii) The *universal* relation on X, is reflexive.

b) *by definition under (a) above, **R is not reflexive** on X, if for **at least one** a ∈ X, we have (a, a) ∉ R.* For example,

i) Let X= N = {1, 2, 3, …}**.** *Define (a, b) ∈ R, if and only if the product (a × b) is an even number.* Then 1×1 =1 is not even, therefore (1, 1) ∉ R. Therefore, R is not reflexive on N. Note that there are other elements like 3, for which (3, 3) ∉ R, but we need not discuss other cases, just one example of (1, 1) ∉ R is sufficient to establish *R as not reflexive*.

ii) The relation *is-brother-of* on the set of *all* human beings is not reflexive, because for a female,

say Sheela, *(Sheela, Sheela) ∉ is-brother-of.*

Note: We may formally define *is-brother-of (x, y) = is-male (x)* and *is-sibling-of (x, y).*

iii)The *empty* relation on a set X ≠ ∅is not reflexive, because for a *∈ X, then (a, a)* ∉∅ (because ∅ has no element).

iv) We may construct our own simple counter-examples., e. g. Let X = {1, 2, 3}, and R= {(1, 1), (1, 2), (1,3), (3,3)}. Then as (2, 2) ∉ R, R is not reflexive.

c) ***Anti-reflexive/ irreflexive:*** *A relation R on set X is said to be **anti-reflexive (or irreflexive)**, if **for every** a ∈ X, we have (a, a) ∉R. For example,* the relation

i) is-less-than (<) on numbers is anti-reflexive, as x < x is false for ***every*** number x, i. e. (x, x) ∉ is-less-than

ii) *is-mother-of* on the set of human beings is anti-reflexive, for every human x, (x, x) ∉ i*s-mother-of*

iii) R= {(2, 3), (3,1), (2, 1)} on the set {1, 2, 3} is anti-reflexive.

Remarks:

a) **Difference between not-reflexive and anti-reflexive**

For a relation R to be ***not-reflexive***, we require for ***some*** a ∈ X, (a, a) ***fails*** to be in R. For a relation R to be ***anti-reflexive,*** we require for ***every*** a ∈ X, (a, a) ***fails*** to be in R.

b) **Every anti-reflexive relation is a not-reflexive relation, but a not-reflexive relation may not be anti-reflexive.**

If R is anti-reflexive on set X, then, by definition, for ***every*** *a* $\in X$, $(a, a) \notin R$ **implies** *for at* ***least one*** *a*, $(a, a) \notin R$ **implies** *R is not reflexive.*

Examples:

i) R= {(2, 2), (3,3), (2, 1)} on the set {1, 2, 3} is a not-reflexive relation as (1, 1) $\notin$ R. But it is not anti-reflexive, as (2, 2) $\in$ R.

ii) the relation is-brother-of on the set of all human beings is not reflexive, because, if x is a female human being, then (x, x) $\notin$is-brother-of. But the relation is not anti-reflexive also as for each male x, we have (x, x) $\in$is-brother-of, contradicting the requirement of anti-reflexivity.

c) **Just by changing the underlying set of a relation, its property may change.**

For example,

i) For the relation ***is-brother-of****, if*

a) Underlying set X= set of all males, then the relation is reflexive

b) Underlying set X= set of all females, then the relation is anti-reflexive

ii) For the relation **R on integers**, such that R (x, y) if and only the product x × y is even, then if

a) *Underlying set X= set of all even integers, then the relation is reflexive*

b) *Underlying set X= set of all integers, then the relation is not reflexive, and also not anti-reflexive*

c) *Underlying set X= set of all odd integers, then the relation is anti-reflexive*

Check Your Progress 1

1) Let R be a relation from X= {1, 2, 3, 4}, to Y = {1, 2, 3, 4}, with

R = {(1, 2), (2, 1), (2, 3), (3, 1), (3, 2), (3, 4), (4, 3)}

i) Give Graphic representation of R

ii) Give Matrix representation of R

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2) For each of the following relations, tell whether it is reflexive, not-reflexive or anti-reflexive

i) The relation of *is-congruent-to* on the set X of all triangles
ii) the relation less-than-or-equal-to (≤) on numbers
iii) The relation less-than (<) on N

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2) **Symmetric, Not-Symmetric, Anti-Symmetric, Asymmetric**

a) A relation R on a set X is ***symmetric*** for a, b ∈ X, *if* (a, b) ∈R, *then* ((b, a) ∈R.

Examples: Each of the following is a *symmetric relation*

i) the relation of equality (=) on N= {1, 2, 3, …}.

ii) The relation of *is-congruent-to* on the set X of all triangles

iii) the relation of *is-sibling-of* on the set of all human beings.

But the relation less-than-or-equal-to (≤) is ***NOT symmetric, though ≤ is reflexive*** on numbers.

b) **By definition (a)** above, a relation R on a set X is ***not symmetric*** if there is ***some pair*** a, b ∈ X, with (a, b) ∈R, but (b, a) ∉R. For showing not-symmetric, we need to show ***just one instance*** violating condition of symmetry.

Examples:

i) The relation R of less-than-or-equal-to (≤) on integers is *NOT symmetric, as (3, 4)* ∈ R*, but (4, 3)*∉ R

ii) Relation R of *is-mother-of* on the set of all human beings is not symmetric

iii) Relation R= {(1, 1), (2, 2), (3, 3), (1,2)} on the set {1, 2, 3} is NOT symmetric, as (1,2)∈ R, but (2, 1) ∉R. We may note that R is reflexive relation

a) **Antisymmetric:** a relation R on a set X is **anti-symmetric,** ***if*** we have (a, b) ∈ R, and (b, a) ∈ R—for two elements a, b ∈ X—***then*** we must have a = b**.**

Examples:

i) The relation of *is-less-than-or-equal-to* on integers is antisymmetric

ii) On the set X = {1, 2, 3}:

a) Each of the relations {(1, 1)}, and {(1, 1), (2, 2)} is both symmetric and antisymmetric.

b) The relation {(1, 2), (1, 3)} is antisymmetric, but not symmetric.

Remarks

a) Difference between not-symmetric and anti-symmetric: For a relation R to be ***not-symmetric,*** we require for ***some pair*** a, b ∈ X, if (a, b) ∈ R, then (b, a) ***fails*** to be in R. For a relation R to be ***anti-symmetric,*** we require for ***every pair*** a, b ∈ X, if (a, b) ∈ R, **and a ≠ b** then (b, a) ***fails*** to be in R.

b) Every anti-symmetric relation R is a not-symmetric relation, but a not-symmetric relation need not be anti-symmetric. ***If R is anti-symmetric on set X***, then, by definition, for ***every pair*** *a, b ∈ X, and* **a ≠ b,** if *(a, b)*∈R, then (b, a) fails to be in R. It **implies** *for at* ***least one pair*** *a, b ∈ X, and* **a ≠ b,** if *(a, b)*∈R, then (b, a) fails to be in R, which **implies R is a not-symmetric**

relation. For the converse, only an example is required: The relation R = {(a, b), (a, c), (c, a)}is not-symmetric as (a, b) ∈ R, but (b, a) fails to be in R. However, R is not anti-symmetric as (a, c) ∈ R, and also (c, a) ∈ R.

Check Your Progress 2

1) For each of the following relations, tell whether it is symmetric, not-symmetric, anti-symmetric

   i) The relation of *is-congruent-to* on the set X of all triangles

   ii) the relation less-than-or-equal-to (≤) on numbers

   iii) The relation *is-brother-of,* if underlying set X= set of all male human beings.

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3) **Transitive, Not-Transitive, Anti-Transitive**

a) A relation R on a set X is ***transitive if*** whenever (a, b) ∈R and also (b, c) ∈R—for elements a, b, c ∈ X—***then*** (a, c) ∈R.

b) From (a) above, a relation R on a set X is ***not transitive if***—for ***some*** a, b, c ∈ X—whenever (a, b) ∈R and also (b, c) ∈R ***then*** (a, c) ***fails*** to be in R.

Examples

i) Each of the relations =, ≤, *and* < is transitive on set of numbers.

ii) The relation *is-congruent-with* on the set of all triangles is transitive.

iii) The relation *is-mother-of* is **not** transitive on the set of all human beings *(because, if a is mother of b, and b is mother of c, then a is grand-mother of c, and a is not mother of c.)*

iv) The relation *is-brother-of* is transitive on the set of all human beings.

For (v) & (vi), on the set X = {1, 2, 3},

v) Each of the relations R_1= {(1, 2), (3, 3), (2, 3), (1, 3)}, and R_2 = {(1, 1), (3, 3)} is transitive,

vi) The relation R_3 = {(1, 2), (3, 3), (2, 3), (2, 2), (2, 1)}is not transitive, because, for (1, 2) ∈R and (2, 3) ∈ R, (1, 3) ∉ R. *The proof is completed*. However, the claim could also be established in another way as follows: (1, 2) ∈ R, (2, 1) ∈ R, but (1, 1) ∉ R.

c) **A relation R on a set X is anti-transitive** if—for ***each*** triple a, b, c ∈ X—whenever (a, b) ∈R and also (b, c) ∈R then (a, c) ***fails*** to be in R.

Examples of anti-transitive:

i) In the plane geometry, the relation—in the set of all straight lines—of being perpendicular ($\perp$) is ***anti-transitive***: because, if for any three straight lines *a, b,* and *c, with a* $\perp$ b, and *b* $\perp$ c, then a $\not\perp$ **c (**a is not perpendicular to c**)**

ii) By the same argument discussed earlier, the relation of *is-mother-of* on the set of human beings is ***anti-transitive.***

iii) On the set X = {1, 2, 3}, the relation R= {(1, 2), (2,3), (3, 1)} is anti-transitive, because, only pairs of the form (a, b) and (b, c) are

a) (1, 2), (2,3), for which (1, 3) fails to be in R

b) (2,3), (3, 1), for which (2, 1) fails to be in R

c) (3, 1), (1, 2), for which (3, 2) fails to be in R.

Remarks:

a) Difference between not-transitive and anti-transitive: For a relation R to be ***not-transitive,*** we require for ***some*** triple a, b, c $\in$ X—***if*** (a, b) $\in$R and also, (b, c) $\in$R ***then*** (a, c) ***fails*** to be in R. For a relation R to be ***anti-transitive,*** we require for ***every*** triple a, b, c $\in$ X—if (a, b) $\in$R and also (b, c) $\in$R then (a, c) ***fails*** to be in R.

b) Every anti-transitive relation is a not-transitive relation, but the converse may not be true. From (a) above, ***if R is anti-transitive, then*** for ***every*** triple a, b, c $\in$ X—if (a, b) $\in$R and also (b, c) $\in$R then (a, c) ***fails*** to be in R. Hence, for relation R, for ***at least one*** triple a, b, c $\in$ X—if (a, b) $\in$R and also (b, c) $\in$R then (a, c) ***fails*** to be in R. And hence R is not-transitive. For the opposite direction, just one example is sufficient. Example: (i) R= {(1, 2), (2, 3), (1, 3), (3, 2))} on the set {1, 2, 3} is a ***not-transitive*** relation as (2, 3)$\in$R , (3,2)$\in$R, but (2, 2)$\notin$ R. But it is ***not anti-transitive***, as (1, 2)$\in$R, (2, 3)$\in$R, and also (1, 3)$\in$R.

4) **A Simple Method of Generating Examples and Counter-Examples**

Start with a small set say***, X= {a, b, c}***, and for the required relation R, take pairs, one by one, of elements of X, which satisfy, or violate the conditions as required for the relation R. For example, if R is required to be (i) reflexive & (ii) transitive, but not symmetric. Take pairs which make the relation reflexive, by including in R the pairs (a, a), (b, b) and (c, c). So far, the relation R is transitive and symmetric. But we require R to be not symmetric. Include, say, only (a, b), but not (b, a), so that

{(a, a), (b, b), (c, c), (a, b)}$\subseteq$R.

As L. H. S. satisfies the required conditions, the process may be terminated.

Care must be taken that while satisfying one condition, some earlier satisfied condition may be violated. Then the process needs to be continued. For example, we wish to have a relation which is reflexive, but not symmetric and not transitive. Then we may continue from {(a, a), (b, b), (c, c), (a, b)} $\subseteq$ R, and add a pair so that transitivity is disturbed, e. g.

{(a, a), (b, b), (c, c), (a, b), **(b, c)**} $\subseteq$R, which satisfies the required conditions. If R is required not to be reflexive, exclude, say (a, a) from R. And so on.

Various combinations of the above-mentioned properties lead to some mathematically interesting and practically useful properties of relations. These interesting and useful combinations of properties include

i) Equivalence of relations, and

ii) Partial ordering of relations.

These two form the subject matter of next two subsections.

Check Your Progress 3

1) For each of the following relations, tell whether it is transitive, not-transitive, anti-transitive.

   i) In the plane geometry, the relation R—in the set of all straight lines—is defined as R (a, b), for straight lines a and b, if and only angle between a and b is 30 degrees.

   ii) The relation less-than-or-equal-to ($\leq$) on numbers

   iii) The relation *is-brother-of,* if underlying set X= set of all human beings

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3.2.3 Equivalence Relations, Equivalence Classes, Partition of A Set

A relation R on a set X is said to be an ***equivalence relation***, if R is reflexive, symmetric and transitive. In more detailed form, R is an equivalence relation on set X, if and only if

(i) R is reflexive on X, i. e. for ***every*** x $\in$ X, we have (x, x) $\in$ R, and

(ii) R is symmetric on X, i. e. for x, y$\in$ X, ***if*** (x, y)$\in$ R, ***then*** (y, x) $\in$ R, and

(iii) R is transitive on X, i. e. for x, y, z$\in$ X, ***if*** (x, y)$\in$ R, ***and***(y, z)$\in$ R, ***then*** (x, z) $\in$ R

Examples

i) The relation of *equality (=)* on the set of numbers, say Q, the set of rational numbers, is an equivalence relation

ii) The relation of '*is-parallel-to*' on the set of all straight lines in a plane, is an equivalence relation

iii) The relation of '*is-congruent-to*' on the set of all triangles in a plane, is an equivalence relation

iv) The relation of '*is-sibling-of*' (male or female having same parents) on the set of human beings, is an equivalence relation.

Not-equivalent relation: A relation R on a set X is ***not an equivalence relation***, if R

i) ***fails to be*** reflexive on X: i. e., *even for one* particular $a \in X$, if $(a, a) \notin R$. For example, for set $X = \{1, 2, 3\}$, the relation $R= \{(2, 2), (3, 3), (2, 3), (3, 2)\}$ is not equivalence relation, as it is not reflexive, because $(1, 1) \notin R$, *despite* $(2, 2), (3, 3) \in R$(this is an example of a relation which is symmetric, and transitive, but which is not reflexive)

Or

ii) ***fails to be*** symmetric on X, i. e. *even for one pair* of elements $a, b \in X$, we have $(a, b) \in R$, but $(b, a) \notin R$. For example, $X = \{1, 2, 3\}$, the relation $R= \{(1, 1), (2, 2), (3, 3), (2, 3)\}$ is not symmetric, and hence not equivalence relation on X, as $(2, 3) \in R$, but $(3, 2) \notin R$. (this is an example of a relation which is reflexive, and transitive, but which is not symmetric)

Or

iii)***fails to be*** transitive on X, i. e. for $a, b, c \in X$, if for *pair* of elements $(a, b) \in R$, and $(b, c) \in R$ but $(a, c) \notin R$. For example, $X = \{1, 2, 3\}$, the relation $R= \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$ is not transitive, and hence not equivalence relation on X, as $(1, 2)$, and $(2, 3) \in R$, but $(1, 3) \notin R$. (this is an example of a relation which is reflexive, and symmetric, but which is not transitive)

Next, we discuss a concept which is quite closely associated with equivalence relation, viz Partition of a set. *The close relation (which we will prove) is that for each equivalence relation R on set X, there is a partition of X, and for every partition of a set, say X there is an equivalence relation on X.*

A. Partition of A Set

For a given set X, a partition of X is a non-empty, finite or infinite, collection $\mathcal{P} = \{X_i\}$, where X_i's are mutually disjoint non-empty subsets of X, with their union equal to whole X. More explicitly, a collection $\mathcal{P} = \{X_i\}$ is a partition of X, if

i) $X_i \neq \emptyset$, for each $X_i \in \mathcal{P}$,

ii) $X_i \cap X_j = \emptyset$, for every distinct pair of X_i **and** X_j in $\mathcal{P}$

iii) $\bigcup_{i \in \mathcal{P}} X_i = X$

A non-empty set may have several distinct partitions. Examples: Let X = {a, b, c}. Then each of the following is a partition of X:

i) {{a}, {b}, {c}}; (ii){{a}, {b, c}} (iii) {{a, b}, {c}} (iv) {{b}, {a, c}} (v) {{a, b, c}}

B. Equivalence Class of an Equivalence Relation

For an equivalence relation R on a set X, the subset [a] of X defined as $[a] = \{x: (x, a) \in R\}$ is called an equivalence class of *a* under the equivalence relation R.

Example: Consider the relation R= {(a, a), (b. b), (b, c), (c, b), (c, c)} on set X= {a, b, c}. The relation is an equivalence relation. Two equivalence classes in X under R are {a} and {b, c}.

C. Every equivalence relation on a set X determines a partition of X. Conversely, if we are given a partition of a non-empty set, then it determines an equivalence relation on X.

Let $\mathcal{P} = \{A_i\}$ be a partition of X. Then we define a relation R on X such that for x, y∈ X, (x, y) ∈ R if and only if x and y belong to the same class A_i. of the partition. Then, to show R is an equivalence relation on X: (I) *Reflexive*: for each x ∈ X, the elements x, and x belong to the same class, hence (x, x) ∈ R. (II) *Symmetry:* if (x, y) ∈ R, i. e., if x and y belong to the same class of partition, then y and x also belong to the same class. Hence, (y, x) ∈ R. *(III) Transitivity:* if (x, y) ∈ R, and (y, z) ∈ R. Then, by definition of R, x and y are in the same class A_i, and y and z are in the same class A_j. By definition of partition, A_i and A_j are either equal or are disjoint. But y belongs to both A_i and A_j. Hence, $A_i = A_j$. Implying x and z belong to the same class of the partition; implying $(x, z) \in R$.

D. Applications of Equivalence Relation and Partition of a Set

Motivation for equivalence relation is that for some specific purpose, many apparently distinct elements behave identically, and hence need not be distinguished from each other. By clubbing all such elements, we reduce complexity for designing solutions of problems concerning the purpose. For example, for designing time-table for a school having 1000 students, 5 classes, say, from VIII to XII, each class having 4 sections, say A, B, C & D, and each section having 50 students. Now time-table need not be designed independently for 1000 distinct students, but needs to be designed for only 20 class-sections
(5 x 4). For the purpose of designing time-table, all the 50 students of a class-section are treated as a single entity, which is an equivalence class under the equivalence relation R with (x, y) ∈ R if and only if students x and y belong to the same class-section. These 20 class-sections determine a partition of the set of all students in the school.

The equivalence relation has useful applications in the design of digital computers and other sequential machines.

Check Your Progress 4

1) Show that the three properties of relations, viz. reflexive, symmetric and transitive are independent of each other, i. e. a relation may be (i) reflexive, but neither symmetric, nor transitive; (ii) symmetric, but neither reflexive, nor transitive; and (iii) transitive, but neither reflexive, nor symmetric.

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2) Find out partition of the set X = {1, 2, 3, 4} given by the equivalence relation

R = {(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (3, 1), (2, 4), (4, 2)}

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3.2.4 Partial Order Relation, Partially Ordered Sets

A relation R on a set X is said to be a ***partial order***, if R is reflexive, *antisymmetric* and transitive. In more detailed form, R is an equivalence relation on set X, if and only if

i) R is reflexive on X, i. e. for ***every*** x ∈ X, we have (x, x) ∈ R, and

ii) R is anti-symmetric on X, i. e. for x, y∈ X, ***if*** (x, y)∈ R, ***and*** (y, x) ∈ R, then x=y, and

iii) R is transitive on X, i. e. for x, y, z∈ X,***if*** (x, y)∈ R, ***and***(y, z)∈ R, ***then*** (x, z) ∈ R

Examples

i) The relation of *less-than-or-equal-to (≤)* on the set of numbers, say Q, the set of rational numbers, is a partial order relation

ii) The relation of *subset-of(⊆) on a set of subsets of a given set, is a partial order.*

iii) The relation of is-divisor-of on N, the set of natural numbers is a partial order.

The relation is called partial order because, some pairs of elements of domain under a partial order may not be related. For example, in (ii) just above, the two subsets {a, b, c} and {b, c, d} are not related to each other under *subset-of(⊆)*